Chapter 4
Financial markets

What is money?

- Definition: stock of assets that is generally accepted in payment for goods or services or in the repayment of debts
- Objects that qualify as money under this definition:

  Currency (bills and coins)

  Checking account deposits (on which you can write a check)

  Perhaps even saving deposits

- Money can take different forms

1. Fiat money: money that has no intrinsic value, only decreed by governments as legal tender

2. Commodity money: for example, gold, silver, or cigarettes in prisons

Semantic traps: money, wealth and income

"Joe has a lot of money"

"Mary is making a lot of money"

- Money is distinct from **wealth** and **income**
  - **Wealth**: **stock** of all resources owned by an individual, including all assets (currency and bank deposits, but also bonds, stocks, houses, land, cars,…)
  - **Income**: **flow** of earnings per unit of time

Measuring money

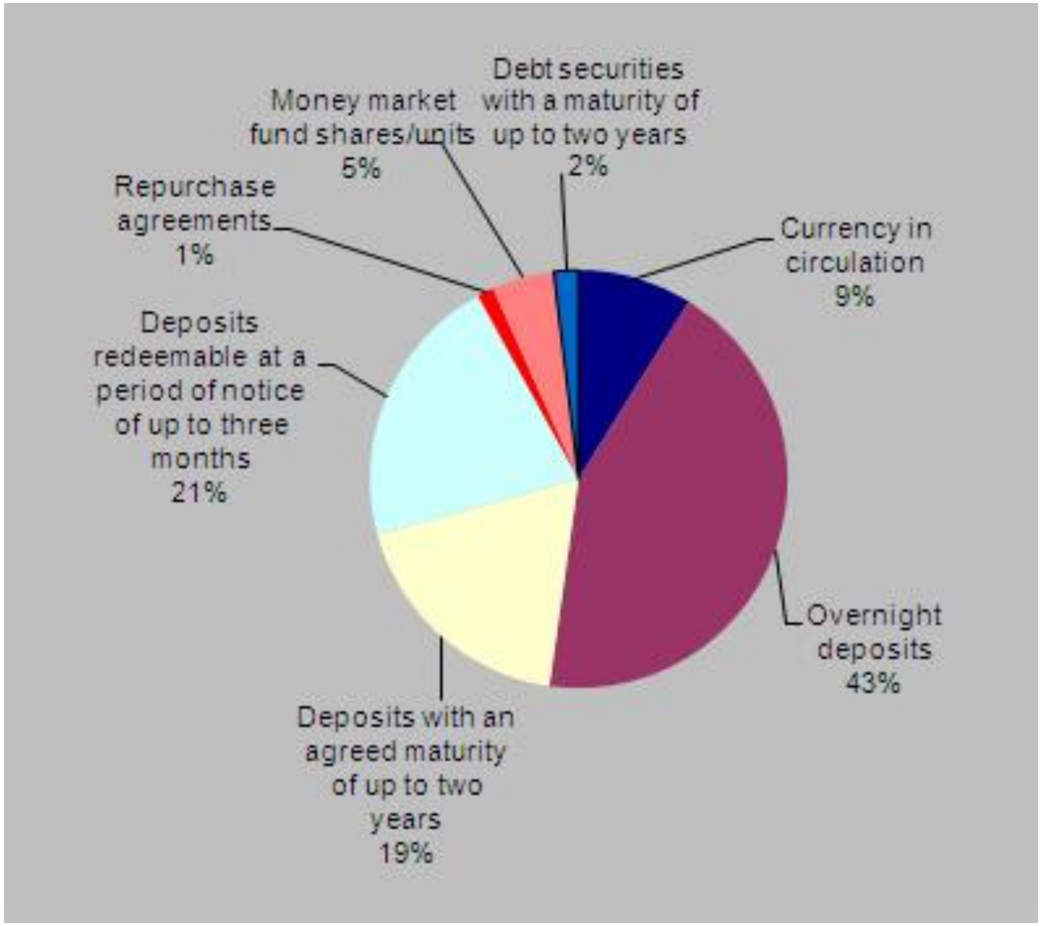
- ECB publishes 3 monetary aggregates
  - M1: currency + checking account deposits (also called overnight deposits)
  - M2: M1 + other short/medium term deposits
  - M3: M2 + marketable instruments (repurchase agreements and money market funds shares) + debt securities up to 2 years

- Fed publishes 2 monetary aggregates
  - M1: currency + checking account deposits (also called demand deposits)
  - M2: M1 + other short/medium term deposits + retail money market funds shares

  (Fed stopped publishing M3 in 2006 as it appeared not to convey any additional information about economic activity that is not already embodied in M2)

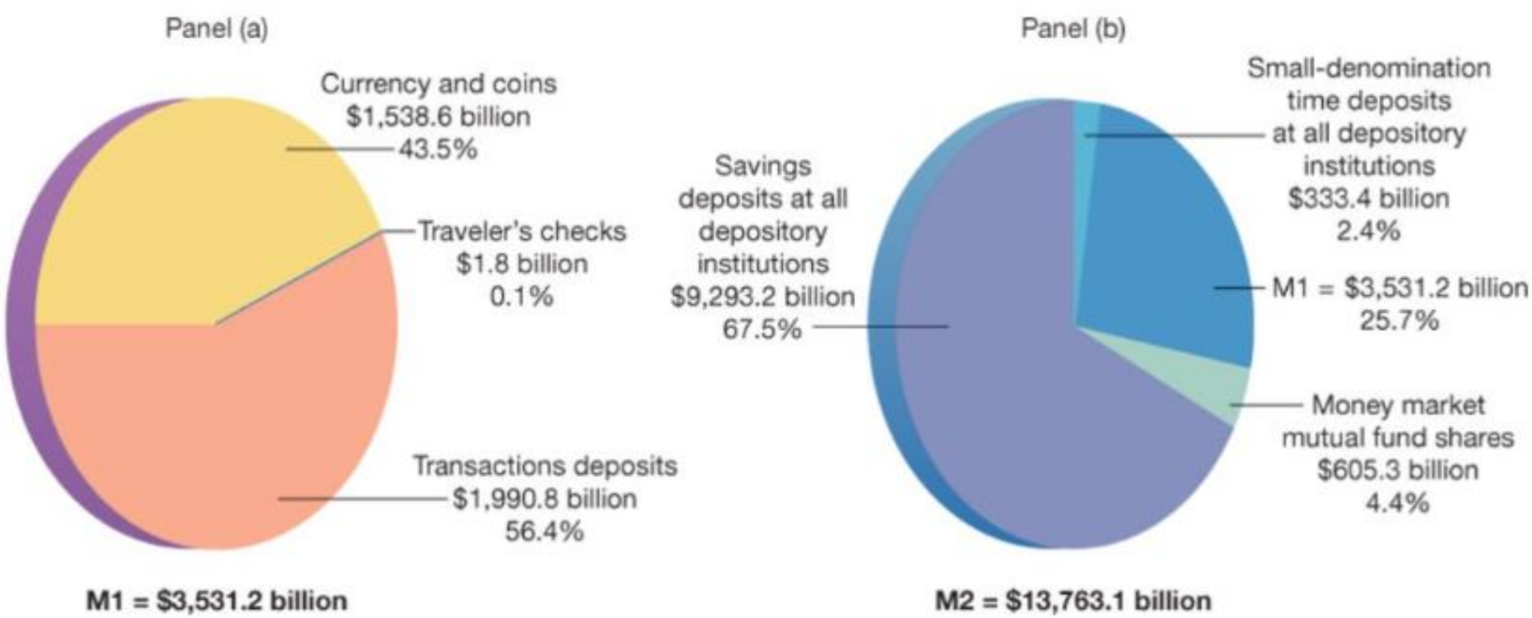
ECB monetary aggregates, 2012

Percentage share of components of M3 at end-2012



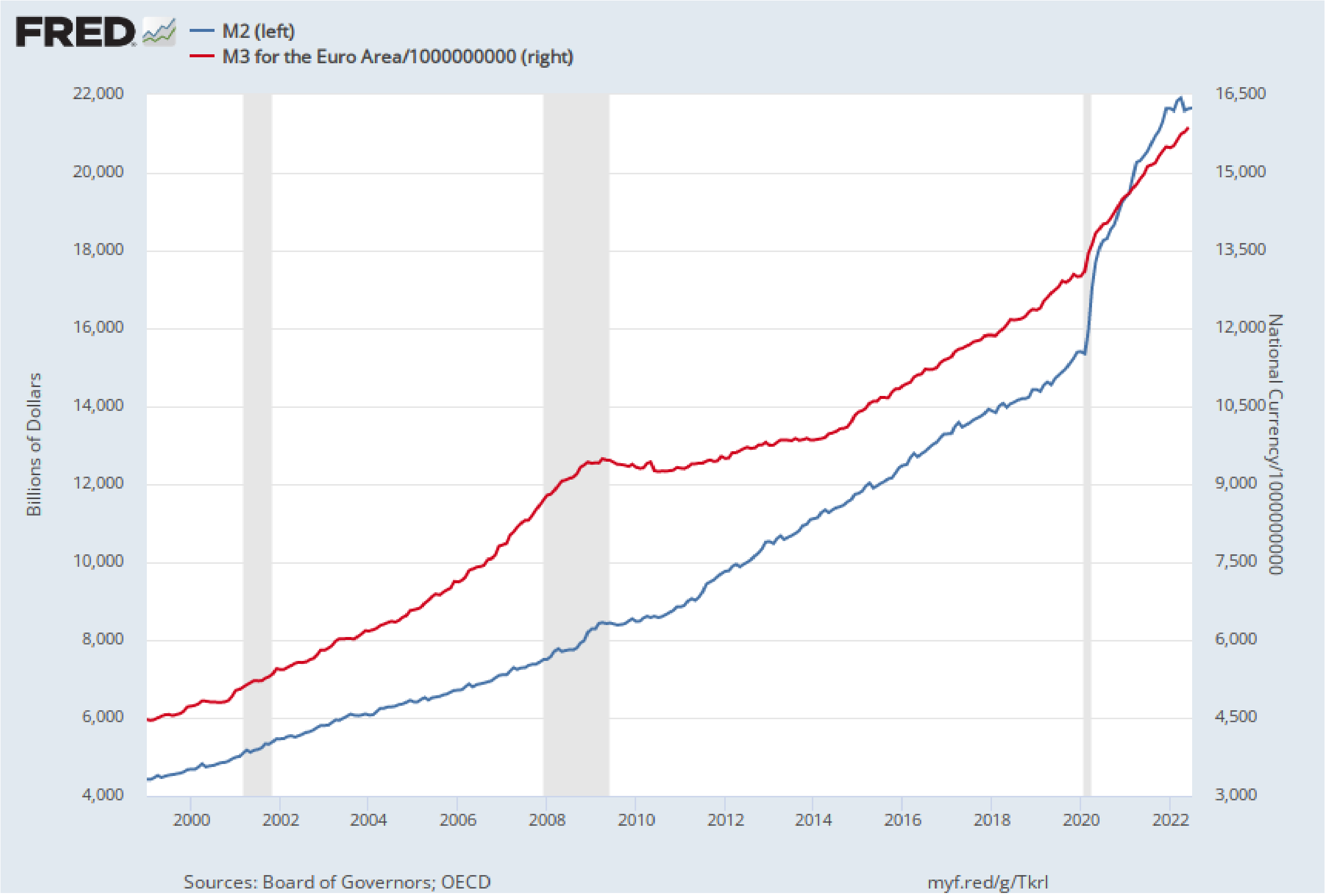
Fed monetary aggregates, 2017

Figure 15-2 Composition of the U.S. M1 and M2 Money Supply, 2017



Panel (a) shows estimates of the M1 money supply, of which the largest component (over 52 percent) is transactions deposits. M2 consists of M1 plus three other components, the most important of which is savings deposits at all depository institutions.

US M2 and EA M3, 1999-2022



Why are monetary aggregates growing over time?

- We will only consider the stricter definition of money

M1 = currency + deposit accounts

- In the **first** part of the class, **only currency**

- In the **second** part of the class, **add deposit accounts** and **banks**

Money and bonds

Consider an economy in which wealth can be hold in only 2 assets

- **Money** can be used for transactions but pays no interest
- **Bonds** cannot be used for transactions but pay a positive interest rate *i*

Wealth = Money + Bonds

For given wealth, a larger demand for money corresponds
to a lower demand for bonds, and vice-versa

- By **holding money**
  - give up interest income, but
  - save on transaction costs (time costs or transaction fees) associated with selling the bonds whenever you need money
- By **holding bonds**
  - gain interest income, but
  - will incur in high transaction costs whenever you need money

Demand for money

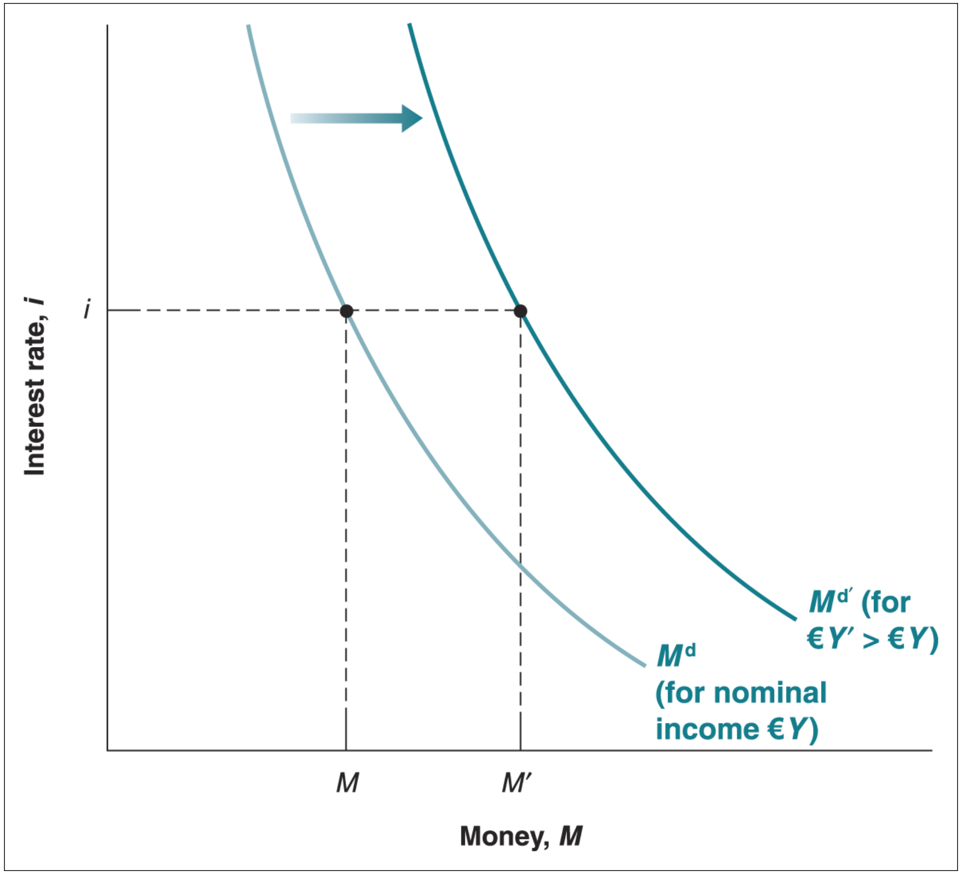
- The demand for money will depend on
  - the **level of transactions**, ***€Y*** (proxied with nominal income)
  - the **interest rate on bonds**, ***i***

- Money demand

$$M^d = €Y \underset{(-)}{L(i)}$$

  - raises proportionally with nominal income
  - decreases with the interest rate on bonds according to the function L

$$M^d = €Y\ \underset{(-)}{L(i)}$$



- For a given level of nominal income, a lower interest rate increases the demand for money (movement along the curve)
- At a given interest rate, an increase in nominal income increases the demand for money (shifts of the curve to the right)

Equilibrium interest rate
Part I: only currency

- Equilibrium in financial markets requires

Money supply = Money demand

$$M^{s} = M^{d}$$

- This equilibrium relation is called the ***LM* relation**

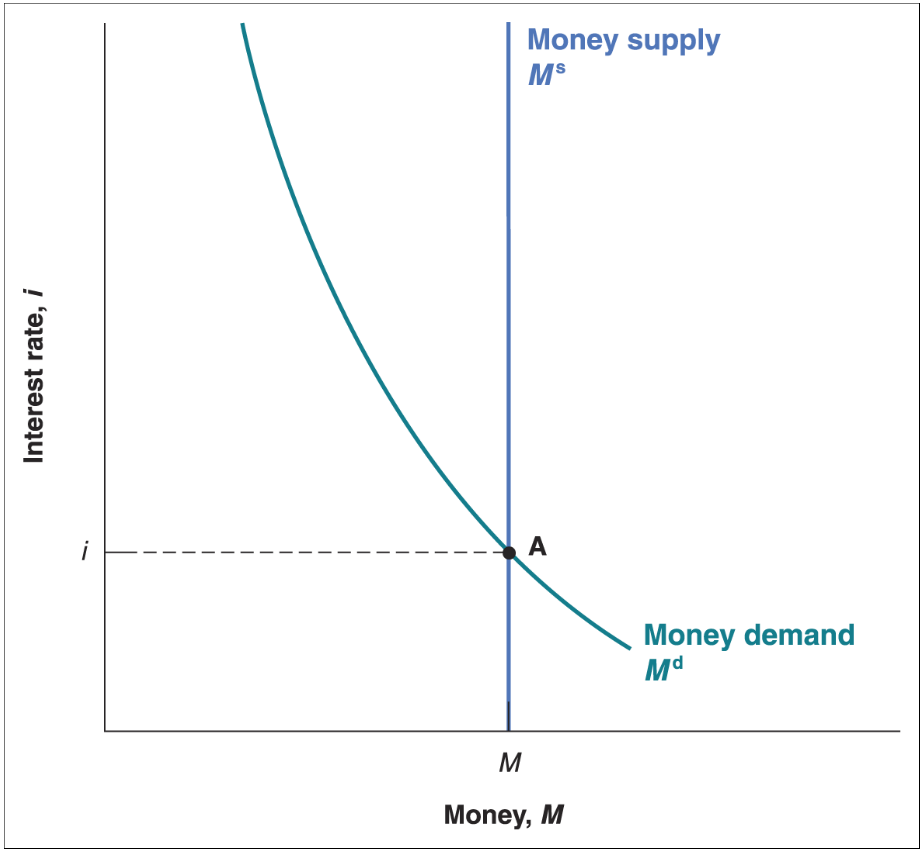
Supply of money

- Money is only currency
- The central bank decides to supply a quantity of money equal to M

$$M^s = M$$

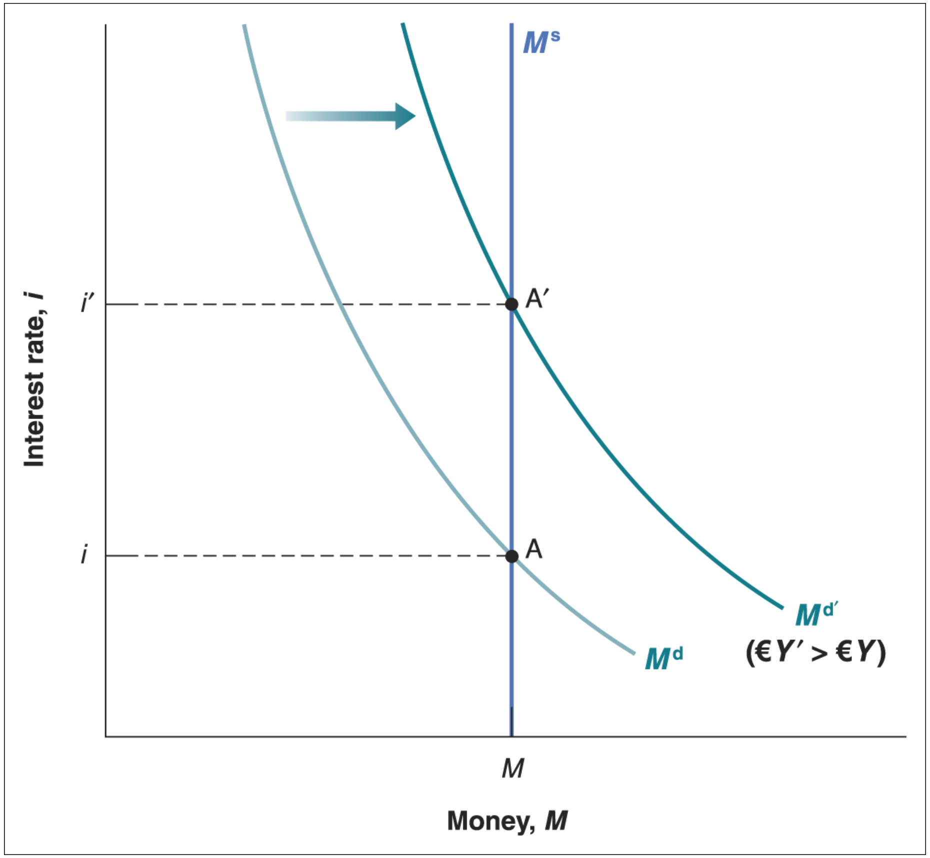
$$M^s = M^d$$

$$M = €Y\,L(i)$$



Equilibrium interest rate must be such that the supply of money (independent of the interest rate) is equal to the demand for money (dependent on the interest rate)

What happens if nominal GDP increases?

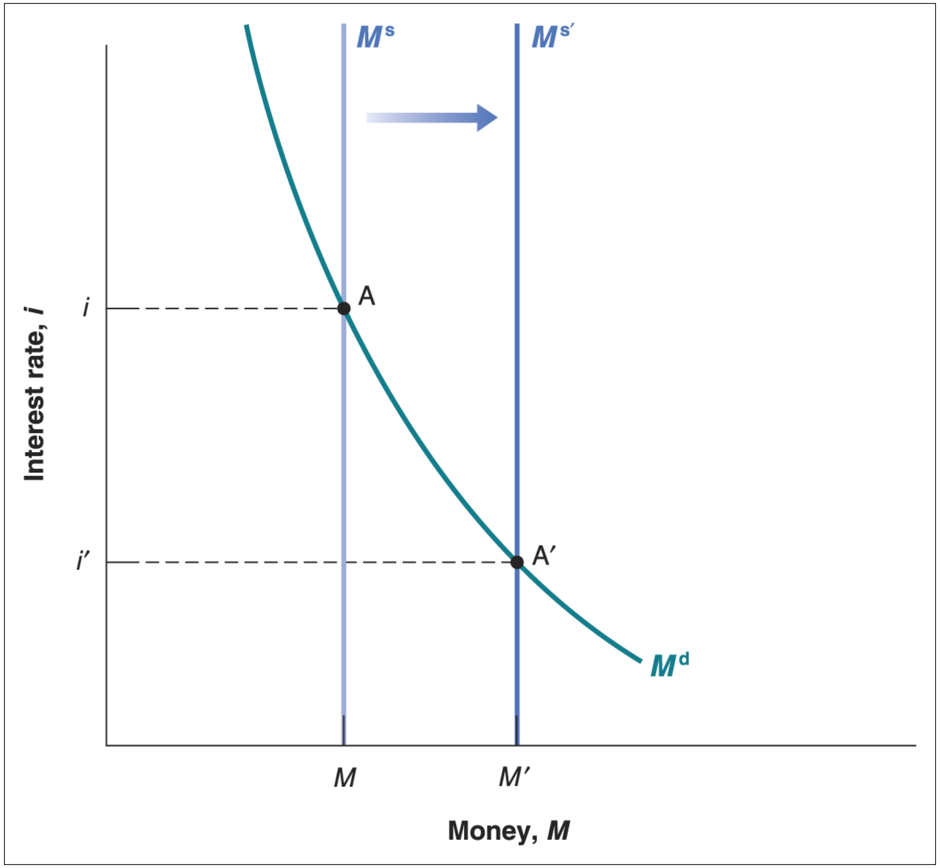


- **Graphical analysis**: At given interest rate, an increase in nominal income leads to an increase in the demand for money shifting the money demand curve to the right. The equilibrium shifts from A to A'.

- **Result**: Given the money supply, an increase in nominal income leads to an increase in the interest rate

- **Economic intuition**: at the initial interest rate, the demand for money exceeds money supply; an increase in the interest rate is needed to decrease the amount of money people want to hold and restore equilibrium

What happens if money supply increases?



- **Graphical analysis**: An increase in money supply shifts the money supply curve to the right. The equilibrium shifts from A to A′.

- **Result**: An increase in money supply leads to a decrease in the interest rate

- **Economic intuition**: the decrease in the interest rate increases the demand for money so it equals the now larger money supply

Monetary policy and open market operations

- **Open-market operations**, which take place in the "open market" for bonds, are the standard method central banks use to change the money stock in modern economies

- If the central bank buys bonds and pays for them by creating money, this operation is called an **expansionary open market operation** because the central bank increases (**expands**) the supply of money

- If the central bank sells bonds and removes from circulation the money it receives in exchange, this operation is called a **contractionary open market operation** because the central bank decreases (**contracts**) the supply of money

CB balance sheet and open market operations

(a) **Balance sheet**

Assets	**Liabilities**
Bonds	Money (currency)

(b) **The effects of an expansionary open market operation**

Assets	**Liabilities**
Change in bond holdings: +€1 million	Change in money stock: +€1 million

Bond prices and bond yields

- Treasury bills or T-bills are issued by the US governments promising payment in a year or less
- If you buy the bond today and hold it for a year, the yield on holding a \$100 T-bill is

$$i = (€100 - €P^b) / €P^b$$

- Alternatively, if you know the yield you can obtain the price

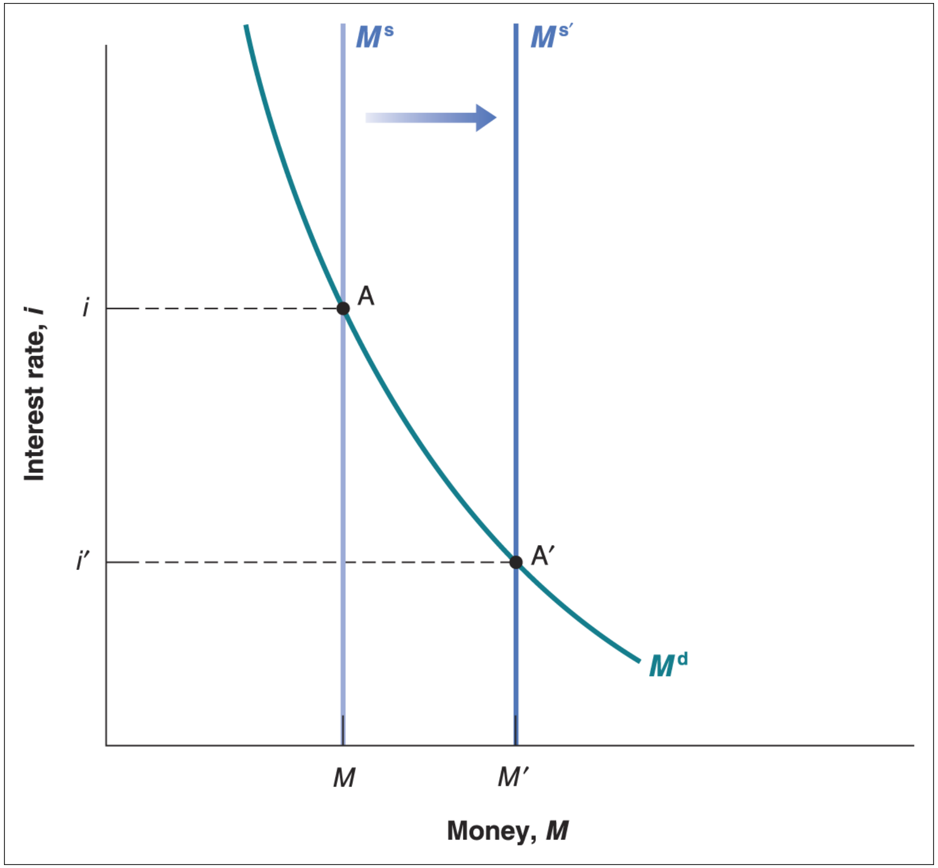
$$€P^b = €100 / (1+i)$$

- There is a negative relation between the price and the yield

What have we learned so far?

- The interest rate is determined by the equality of the supply of money and the demand for money
- By changing the supply of money, the central bank can affect the interest rate
- The central bank changes the supply of money through open market operations, which are purchases or sales of bonds in exchange of money
- Open market operations in which the central bank increases the money supply by buying bonds lead to an increase in the price of bonds and a decrease in the interest rate
- Open market operations in which the central bank decreases the money supply by selling bonds lead to a decrease in the price of bonds and an increase in the interest rate

Choosing money or choosing the interest rate?



- A decision by the central bank to lower the interest rate from *i* to *i* ’ is equivalent to increasing the money supply from *M* to *M’*
- Rather than the money supply, the central bank could have chosen the interest rate and then adjusted the money supply so as to achieve the interest rate it had chosen
- “Choosing” the interest rate, instead of money supply, is what modern central banks, including the Fed and the ECB, typically do

A monetary policy decision

Federal Reserve press release, January 29, 2025

Recent indicators suggest that economic activity has continued to expand at a solid pace. The unemployment rate has stabilized at a low level in recent months, and labor market conditions remain solid. Inflation remains somewhat elevated.

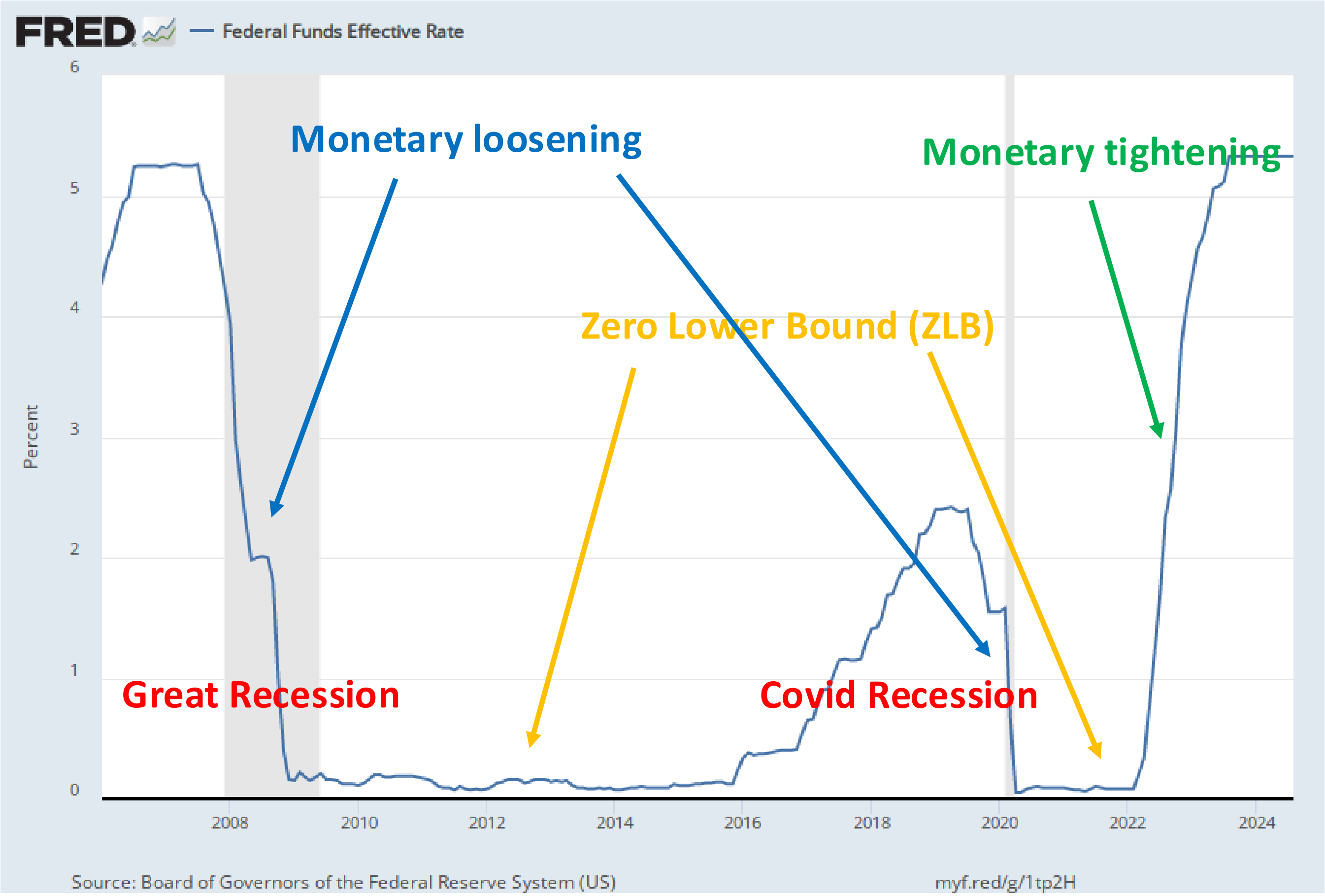
The Committee seeks to achieve maximum employment and inflation at the rate of 2 percent over the longer run. The Committee judges that the risks to achieving its employment and inflation goals are roughly in balance. The economic outlook is uncertain, and the Committee is attentive to the risks to both sides of its dual mandate.

In support of its goals, the Committee decided to maintain the target range for the federal funds rate at 4-1/4 to 4-1/2 percent. In considering the extent and timing of additional adjustments to the target range for the federal funds rate, the Committee will carefully assess incoming data, the evolving outlook, and the balance of risks. The Committee will continue reducing its holdings of Treasury securities and agency debt and agency mortgage-backed securities. The Committee is strongly committed to supporting maximum employment and returning inflation to its 2 percent objective.

In assessing the appropriate stance of monetary policy, the Committee will continue to monitor the implications of incoming information for the economic outlook. The Committee would be prepared to adjust the stance of monetary policy as appropriate if risks emerge that could impede the attainment of the Committee’s goals. The Committee’s assessments will take into account a wide range of information, including readings on labor market conditions, inflation pressures and inflation expectations, and financial and international development.

Voting for the monetary policy action were Jerome H. Powell, Chair; John C. Williams, Vice Chair; Michael S. Barr; Michelle W. Bowman; Susan M. Collins; Lisa D. Cook; Austan D. Goolsbee; Philip N. Jefferson; Adriana D. Kugler; Alberto G. Musalem; Jeffrey R. Schmid; and Christopher J. Waller.

Federal Funds Rate, U.S. 2006-2024



During both the Great Recession and the Covid recession, the FFR reached the ZLB

Since March 2022, the Fed has been tightening monetary policy

Now we introduce

- **checking account deposits,** so that money = currency + deposit accounts
- **banks** supplying deposits accounts

Banks

Banks receive funds from people and firms in the form of deposit accounts, and use these funds to buy bonds or to make loans to other people and firms

- Banks receive funds from people and firms: the **liabilities** of the banks are therefore equal to the value of these **checkable deposits**

- Banks keep as **reserves** some of the funds they receive (either in the form of **cash** or in an **account that banks have a central banks**)

- The **assets** of banks are the **bonds** and the **loans** plus the **reserves**

Balance sheet of banks

Assets	Liabilities
Bonds	Deposits accounts
Loans	
Reserves	

Banks hold **reserves** for three reasons:

1. On any given day, some depositors withdraw cash from their checking accounts, while others deposit cash into their accounts with no reason for these flows to balance → banks need to keep reserves

2. On any given day, people with accounts at the bank write checks to people with accounts at other banks, and people with accounts at other banks write checks to people with accounts at the bank → banks need to keep reserves

3. Banks are typically subject to **reserve requirements**, which require them to hold reserves in some proportion of their deposits

Balance sheets of central bank and banks

(a) **Central Bank**

Assets	**Liabilities**
Bonds	Central bank money = Reserves + Currency

(b) **Banks**

Assets	**Liabilities**
Reserves Loans Bonds	Deposit accounts

Equilibrium interest rate
Part II: adding deposits accounts and banks

- Equilibrium in financial markets requires

Supply of CB money = Demand of CB money

$$H^s = H^d$$

- Supply of central bank money H^s: under the direct control of the CB
- Demand of central bank money H^d?
  - Demand for currency by people, CU^d
  - Demand for reserves by banks, R^d

Demand for money, currency and deposits

People can now hold both **currency** and **checkable deposits**

Therefore, **two** decisions:

- First, people must decide how much money to hold. As before

$$M^d = €Y\, L(i)$$

- Second, they must decide how much of this money to hold in currency and how much to hold in checkable deposits

$$CU^d = c\, M^d$$

$$D^d = (1-c)\, M^d$$

that is, people hold a fixed proportion c of their money in currency and a fixed proportion 1-c in deposits

Demand for reserves

- Banks holds a fraction of deposits as reserves
- We know D^d from the demand for deposit accounts
- Demand for reserves is

$$R^d = \theta D^d = \theta(1-c)M^d$$

where θ is the fraction of reserves to deposits

Demand for central bank money

H^d = currency by people + reserves by banks

$$H^d = CU^d + R^d$$

$$= cM^d + \theta(1-c)M^d$$

$$= [c + \theta(1-c)]M^d$$

$$= [c + \theta(1-c)]€YL(i)$$

- Both the demand for currency and the demand for reserves are proportional to the demand for money
- Hence, they increase with *€Y* and decrease with the interest rate *i*

Supply of CB money

- CB money is currency + reserves
- The central bank decides to supply a quantity of CB money equal to H

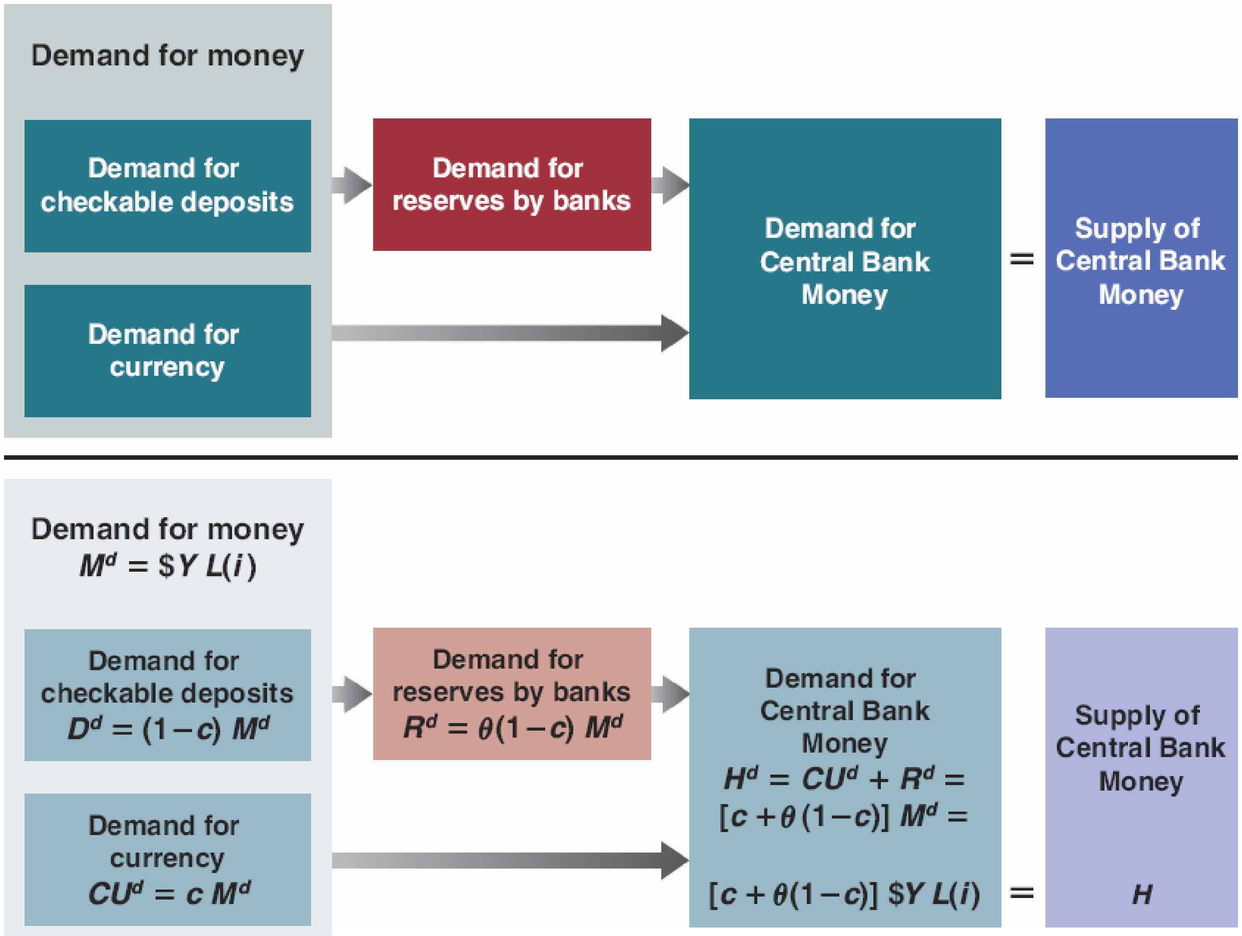
$$H^s = H$$

Supply of CB money = Demand of CB money

$$H^{s} = H^{d}$$

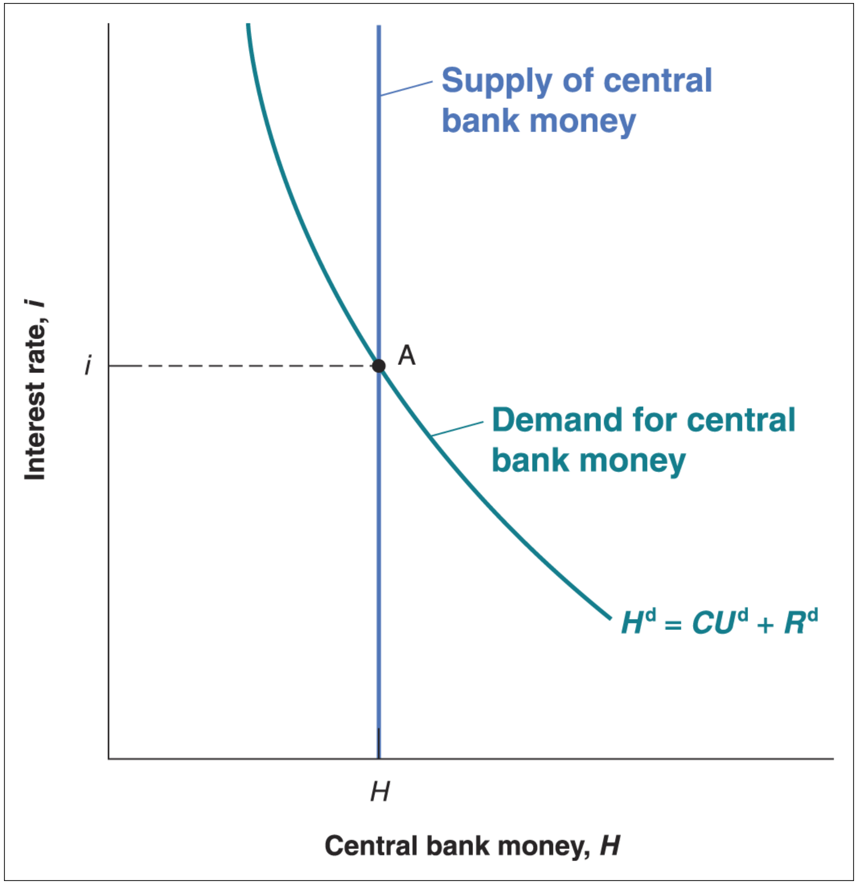
$$H = [c + \theta(1-c)]€YL(i)$$

Determinants of demand and supply of central bank money



Note: In the textbook, to simplify the analysis, it is assumed that c=0 or CU^d=0
That is, people only hold money in the form of deposit accounts

With a graph…



- The supply of CB money is under control of the CB and independent of the interest rate
- The demand of CB money depends negatively on the interest rate for 2 reasons
- First, an increase in the interest rate decreases the demand for money and thus the demand for currency
- Second, as the demand for money decreases also the demand for deposits goes down and this decreases the demand for reserves from banks
- The effects of an increase in nominal income or in the supply of CB money are qualitatively the same as before

Same as before!

2 alternative ways to look at the equilibrium

- We have seen: Supply of CB of money = Demand of CB money
- But we can represent the equilibrium also as
    - Money supply = money demand
    - Supply of reserves = demand of reserves

Money supply = money demand

- Start from the condition that $H^s = H^d$

$$H = [c + q(1- c)]\$YL(i)$$

- Divide both sides by [c + θ(1-c)] to get

$$\underbrace{\frac{1}{[c+\theta(1-c)]}H}_{\text{Money supply}} = \underbrace{\$YL(i)}_{\text{Money demand}}$$

- The overall supply of money is equal to central bank money H times 1/[c + θ (1-c)]

Supply of reserves = demand for reserves

- Start from the condition that $H^s = H^d$

$$H = CU^d + R^d$$

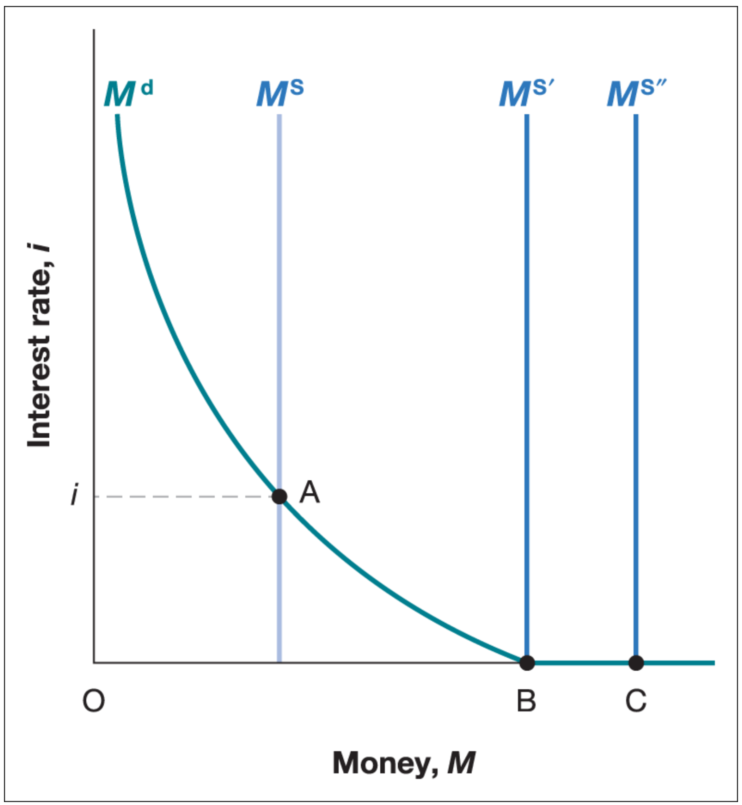
- Bring CU^d on the other side to get

$$\underbrace{H - CU^d}_{\text{Supply of reserves}} = \underbrace{R^d}_{\text{Demand for reserves}}$$

- **Federal funds market** is an actual market for bank reserves in the US
- **Federal funds rate** is the interest rate determined on that market
- Federal funds rate is the **main indicator of US monetary policy** since the Fed can choose the federal funds rate it wants by changing H

Powerless monetary policy: the liquidity trap

What happens once the interest rate falls to the **zero lower bound**?



For now, ignore banks and assume all money is currency

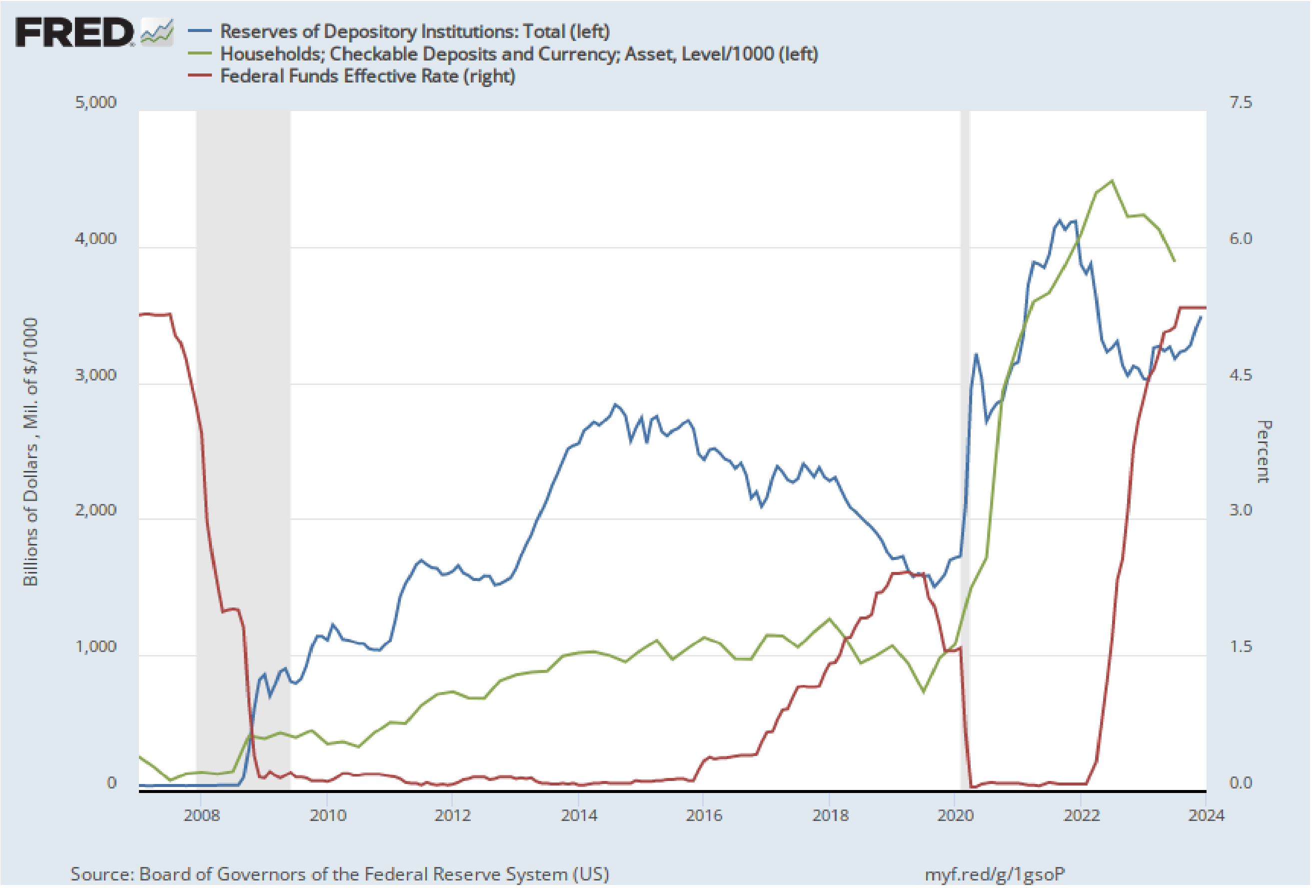
When the interest rate reaches 0, and once people have enough money for transactions, they become indifferent between holding money and holding bonds → the demand for money becomes horizontal

As the money supply increases, individuals are willing to hold more money at the same interest rate, namely zero

This implies that, when the interest rate equals 0, further increases in money supply have no effect on the interest rate

- What happens if we also consider banks and checkable deposits?
- A similar argument as for the demand for money by people also applies to the demand for reserves by banks
- If the interest rate is equal to 0, banks are indifferent between holding reserves and buying bonds
- Thus, when the interest rate is at the zero lower bound, and the central bank increases the money supply, we are likely to see an increase in checkable deposits + currency and bank reserves, with the interest rate remaining at 0

The liquidity trap or zero lower bound in action



The large increase in the supply of central bank money caused by the Great Recession and the Covid Recession was absorbed by households and banks